

Finding All Permutations Without Repetition: A Wedding Seating Chart

Imagine you must arrange the seating chart for a wedding reception with delicate social requirements. Some of the guests hate each other, while others demand to sit near an influential guest. The seats at the rectangular table form one long, straight row, rather than a circle. It'd be helpful for your planning to see every possible ordering of guests—that is, every permutation without repetition of the set of guests. No repetition occurs, because each guest appears in the seating chart only once.

Let's use a simple example of Alice, Bob, and Carol, or $\{A, B, C\}$. [Figure 6-2](#) shows all six possible permutations of these three wedding guests.

One way we can determine the number of permutations without repetition is with a head-tail recursive strategy. We select one element from the set as the head. We then get every permutation of the rest of the elements (which constitute the tail), and for each permutation we place the head in every possible location in the permutation.

In our ABC example, we'll start with Alice (A) as the head and Bob and Carol (BC) as the tail. The permutations of $\{B, C\}$ are BC and CB. (How we got BC and CB is explained in the next paragraph, so just put that question aside for now.) We'll put A in every possible location in BC. That is, we put Alice before Bob (ABC), in between Bob and Carol (BAC), and after Carol (BCA). This creates the permutations ABC, BAC, and BCA. We also put A in every possible position in CB, creating ACB, CAB, and CBA. This creates all six permutations of Alice, Bob, and Carol sitting at the reception table. Now we can pick the arrangement that results in the fewest fights (or the most fights, if you want a memorable wedding reception).